NEUROLOGICAL DISORDERS IN PREGNANCY

1. OVERVIEW

General Concept

- Neurological disorders may either pre-exist before pregnancy or develop newly during pregnancy, requiring careful differentiation from pregnancy-related complications.
- Most women with neurological disorders have favorable outcomes, but certain conditions significantly increase maternal risk.
- Neurovascular disorders contribute to more than 7% of maternal deaths, making them clinically important.
- Psychiatric disorders can present with neurological-like symptoms such as cognitive and neuromuscular abnormalities, complicating diagnosis.

2. CENTRAL NERVOUS SYSTEM IMAGING

Role of Imaging

- Computed tomography (CT) and magnetic resonance imaging (MRI) are essential tools for diagnosis, classification, and management of neurological disorders.
- CT is preferred in emergency situations because it rapidly detects acute hemorrhage.
- MRI is preferred for detailed evaluation because it better detects demyelinating diseases, AV malformations, congenital abnormalities, and spinal cord pathologies.

Safety and Technique

- Imaging can be safely performed during pregnancy with proper precautions.
- Pregnant women in advanced stages should be positioned in lateral tilt to prevent hypotension and reduce imaging artifacts caused by aortic pulsations.

Advanced Imaging

- Cerebral angiography helps diagnose and treat cerebrovascular diseases but involves radiation exposure, requiring shielding.
- PET and fMRI are not well studied in pregnancy and are not routinely recommended.

3. HEADACHE DISORDERS

Epidemiology and Importance

- Headaches are very common in women of reproductive age and frequently occur during pregnancy.
- Most headaches in pregnancy are primary, with migraine being the most common diagnosis.
- Secondary headaches are often related to hypertensive disorders, making them clinically important.

Headache Classification (Table Explanation)

Primary Headaches

- Primary headaches include migraine, tension-type headache, trigeminal cephalalgia, and other idiopathic forms.
- These occur without an underlying structural or systemic cause.

Secondary Headaches

- Secondary headaches result from identifiable causes such as trauma, vascular disorders, infections, substance abuse, metabolic disturbances, craniofacial disorders, and psychiatric conditions.

IMPORTANT

- Primary headaches are more common, but secondary headaches are more dangerous and require urgent evaluation.

Causes Distribution (Figure Explanation)

- The pie chart shows that migraine accounts for approximately 59% of headaches in pregnancy, making it the dominant cause.
- Hypertensive disorders contribute around 18%, highlighting their clinical significance.
- Other causes include infections, stroke, pituitary disorders, and miscellaneous conditions, each contributing smaller percentages.

HIGH-YIELD

- Migraine is the most common cause, but hypertensive causes are the most dangerous.

4. TENSION HEADACHE

Clinical Features

- Tension headache is the most common type of headache and is usually mild to moderate in intensity.
- Pain is typically located in the back of the head and neck and is associated with muscle tightness.
- Neurological symptoms such as nausea or aura are usually absent, which helps differentiate it from migraine.

Management

- Symptoms usually improve with rest, massage, heat therapy, NSAIDs, or mild sedatives.
- Chronic cases may respond to medications like amitriptyline.

5. MIGRAINE

Definition and Types

- Migraine is a recurrent neurological disorder characterized by episodic severe headache with autonomic symptoms.
- It is classified into migraine without aura, migraine with aura, and chronic migraine.

Types Explained

- Migraine without aura presents as unilateral throbbing headache with nausea, vomiting, and photophobia.

- Migraine with aura includes premonitory neurological symptoms such as visual disturbances before headache onset.
- Chronic migraine is defined as headache occurring at least 15 days per month for more than 3 months.

Pathophysiology (Logical Flow)

- Neuronal dysfunction leads to decreased cortical blood flow.
- This triggers activation of vascular and meningeal pain receptors.
- Trigeminal sensory pathways become stimulated.
- These processes collectively result in migraine pain and associated symptoms.

Clinical Relevance

- Migraine is strongly influenced by hormonal changes, explaining its prevalence in women.
- Migraines with aura are associated with increased risk of ischemic stroke.
- Associated conditions include anxiety, depression, asthma, and chronic pain syndromes.

Migraine in Pregnancy

- Most women experience improvement in migraine frequency during pregnancy.
- New-onset migraines, especially with neurological symptoms, require full evaluation to exclude serious causes.

CLINICAL ALERT

- Migraine during pregnancy is associated with increased risk of:
    - Preeclampsia
    - Gestational hypertension
    - Preterm birth
    - Ischemic stroke

Management of Migraine

First-Line Treatment

- Mild headaches are treated with acetaminophen or NSAIDs along with antiemetics.

Severe Cases

- Severe migraines require parenteral therapy including IV fluids, antiemetics, and triptans.
- Opioids are reserved only for refractory cases due to risk of dependency.

Drug Considerations (Table Explanation)

- Simple analgesics include acetaminophen and NSAIDs for mild pain relief.
- Triptans act as serotonin receptor agonists and are effective in reducing headache and associated symptoms.
- Dopamine antagonists like metoclopramide help control nausea and vomiting.
- Antiemetics such as ondansetron are used for symptom relief.

IMPORTANT

- Ergot derivatives are contraindicated in pregnancy because they cause uterine contractions.
- Chronic NSAID use should be avoided in late pregnancy due to risk of fetal complications.

Preventive Therapy

- Frequent migraines require prophylactic therapy using drugs such as amitriptyline, propranolol, or metoprolol.
- Newer therapies like CGRP inhibitors and botulinum toxin are not well studied in pregnancy.

6. CLUSTER HEADACHE

Features

- Cluster headache is a rare but severe primary headache disorder characterized by unilateral intense pain around the eye.
- Attacks last 15–180 minutes and are associated with autonomic symptoms such as tearing and nasal congestion.

Management

- Acute treatment includes 100% oxygen therapy and subcutaneous sumatriptan.
- Prophylaxis is typically done with verapamil.

7. SEIZURE DISORDERS (EPILEPSY)

Definition

- A seizure is a transient episode caused by abnormal excessive neuronal activity in the brain.

Causes

- Causes include trauma, infections, tumors, metabolic abnormalities, substance withdrawal, and vascular malformations.
- Idiopathic epilepsy is diagnosed only after excluding identifiable causes.

Classification (Stepwise Understanding)

Focal Seizures

- Originate in a specific brain region and affect localized neurological functions.
- Can occur with preserved awareness or impaired awareness.

Generalized Seizures

- Involve both hemispheres and lead to loss of consciousness.
- Include tonic-clonic seizures with muscle rigidity followed by rhythmic jerking.

Absence Seizures

- Characterized by brief loss of consciousness without motor activity and rapid recovery.

Unknown Onset

- Occur when the origin is unclear and later classified based on further information.

8. PRECONCEPTION COUNSELING

- Women with epilepsy should receive counseling before pregnancy.
- Folic acid supplementation should begin before conception and be increased during pregnancy.
- Medication should be optimized to use the lowest effective dose and preferably monotherapy.
- Drug withdrawal may be considered if the patient has been seizure-free for at least 2 years.

9. PREGNANCY COMPLICATIONS IN EPILEPSY

Maternal Risks

- Increased seizure frequency may occur due to physiological changes affecting drug levels.
- Status epilepticus is a life-threatening emergency requiring immediate treatment.

Fetal Risks

- Antiepileptic drugs increase the risk of congenital malformations.
- Risk is higher with polytherapy compared to monotherapy.

Drug Teratogenicity (Table Explanation)

- Valproate has the highest teratogenic risk, causing neural tube defects and developmental delay.
- Phenytoin causes fetal hydantoin syndrome with craniofacial abnormalities.
- Carbamazepine and phenobarbital also increase malformation risk.
- Lamotrigine and levetiracetam have relatively lower risk compared to other drugs.

IMPORTANT

- Risk of malformations increases with:
    - Higher drug dose
    - Multiple drug therapy

10. MANAGEMENT OF EPILEPSY IN PREGNANCY

Treatment Principles

- The primary goal is seizure prevention to reduce maternal and fetal complications.
- Use the minimum number of drugs at the lowest effective dose.
- Avoid seizure triggers such as sleep deprivation and stress.

Monitoring

- Routine drug level monitoring may not always be reliable due to physiological changes in pregnancy.
- Drug levels are most useful when assessing compliance or after seizure episodes.

Fetal Monitoring

- Targeted ultrasound is recommended to detect congenital anomalies.
- Routine fetal surveillance is not required in uncomplicated epilepsy.

Additional Considerations

- Breastfeeding is generally safe as no major adverse cognitive effects have been observed.
- Hormonal contraceptives may be less effective with certain antiepileptic drugs, requiring alternative methods.

11. CEREBROVASCULAR DISEASES

Overview

- Includes ischemic stroke, hemorrhagic stroke, and vascular malformations such as aneurysms and AVMs.
- Pregnancy increases both immediate and long-term risk of stroke.

Pathophysiology

- Ischemic stroke occurs due to reduced blood flow leading to brain tissue infarction.
- Hemorrhagic stroke occurs due to bleeding into brain tissue, causing pressure and toxic effects.

Epidemiology

- Stroke incidence in pregnancy ranges from 10 to 40 per 100,000 births.
- Although rare, stroke contributes significantly to maternal mortality.

Risk Factors

Non-Pregnancy Related

- Advanced age, hypertension, obesity, diabetes, smoking, and cardiac disease increase stroke risk.

Pregnancy-Related

- Hypertensive disorders are the most common cause of stroke in pregnancy.
- Other factors include gestational diabetes, hemorrhage, cesarean delivery, and sepsis.

HIGH-YIELD

- Pregnancy-associated hypertension is the strongest risk factor for stroke.

Hemodynamic Changes

- Pregnancy causes increased autoregulation of cerebral blood flow.
- Cerebral blood flow decreases by about 20% during mid-pregnancy but increases in hypertensive states.

12. ISCHEMIC STROKE

Mechanism (Flow)

- Occlusion or embolism blocks cerebral blood vessels.
- Blood supply to brain tissue decreases.
- Tissue ischemia occurs, leading to infarction and cell death.

TIA (Transient Ischemic Attack)

- TIA is caused by temporary ischemia and symptoms resolve within 24 hours.
- Approximately 10% of patients develop stroke within one year.

Causes (Table Explanation)

Ischemic Stroke Causes

- Includes preeclampsia, arterial or venous thrombosis, antiphospholipid syndrome, migraine, embolism, and vasculitis.

Hemorrhagic Stroke Causes

- Includes chronic hypertension, aneurysm rupture, AV malformations, vasculopathy, and substance abuse.

Clinical Features

- Sudden onset severe headache is a common presenting symptom.
- Neurological deficits such as hemiplegia and seizures may occur.

Diagnosis

- Immediate CT scan is required for rapid diagnosis.
- Further evaluation includes MRI, angiography, echocardiography, and blood tests for thrombophilia.

Management

- Treatment includes identifying the cause and restoring blood flow.
- Thrombolysis may be considered in selected cases.
- Mechanical thrombectomy is an option in severe cases.

13. DIAGRAM EXPLANATION (BRAIN STROKE FIGURE)

- The diagram illustrates different stroke types including:
    - Subcortical infarction caused by reduced blood supply.
    - Hypertensive hemorrhage due to vessel rupture.
    - Aneurysm showing vessel wall dilation.
    - Embolism or thrombosis blocking arteries.
    - Arteriovenous malformation causing abnormal vessel connections.
    - Cortical vein thrombosis affecting venous drainage.

CLINICAL MEANING

- Different mechanisms lead to stroke, and identifying the type is essential for treatment.

- Neurological disorders in pregnancy can be pre-existing or new-onset and require differentiation.
- MRI is preferred imaging modality, while CT is used in emergencies.
- Migraine is the most common headache, but hypertensive causes are the most dangerous.
- Migraine with aura increases stroke risk.
- Avoid ergot drugs and chronic NSAIDs in pregnancy.
- Epilepsy requires preconception planning and monotherapy.
- Valproate has highest teratogenic risk, while lamotrigine is relatively safer.
- Seizure control is the primary goal during pregnancy.
- Stroke risk increases in pregnancy, especially with hypertension.
- Ischemic stroke is caused by vessel blockage, while hemorrhagic stroke is due to bleeding.
- TIA is reversible but predicts future stroke risk.
- CT scan is first-line for stroke diagnosis.
- Hypertensive disorders are the most important risk factor for stroke.
- Postpartum period carries high stroke risk.

- Early diagnosis and management significantly reduce maternal mortality.